

Automobile Systems & Electric Vehicles: A Compact Overview

1. Automobile Systems (ICE Vehicle)

An automobile integrates multiple mechanical and electrical systems:

- Engine – Converts fuel into motion via combustion.
- Fuel System – Stores & supplies fuel (tank, pump, injectors).
- Exhaust System – Removes gases & reduces noise.
- Cooling & Lubrication – Prevents overheating & friction.
- Transmission – Transfers power to wheels (gearbox, clutch, differential).
- Braking & Steering – Controls speed & direction.
- Electrical System – Battery, alternator, starter, lights.

These systems work together but require frequent maintenance (oil, filters, belts).

2. EV Components & Working

EVs replace the engine and fuel with an electric motor & battery.

Main Components:

- Traction Battery (Lithium-ion) – Stores energy.
- Electric Motor – Drives wheels.
- Inverter – Converts DC to AC.
- Onboard Charger – Charges battery from grid.
- Regenerative Braking – Recovers energy while braking.

Working:

1. Battery → Inverter → Motor → Wheels.
2. During braking → Motor acts as generator → Battery recharges.

No fuel, no exhaust, no clutch, no gearbox (in most EVs).

3. ICE vs EV – Quick Comparison

Parameter	ICE Vehicle	Electric Vehicle
Energy	Petrol/Diesel	Electricity
Efficiency	~20–30%	~70–85%
Fuel/Charge time	2–5 min	30 min – 8 hrs
Maintenance	High	Low
Tailpipe emission	Yes	No
Range	400–600 km	200–500 km
Upfront cost	Lower	Higher
Running cost	Higher	Lower

4. Advantages & Challenges of EVs

Advantages

- Zero tailpipe pollution
- Low running & maintenance cost
- Silent & smooth drive
- Instant torque (fast acceleration)
- Home charging possible

Challenges

- High initial purchase price
- Limited charging stations
- Longer “refueling” time
- Battery degrades over time
- Disposal/recycling concerns

Summary Table – Advantages & Challenges at a Glance

Category	Details
Advantages	Eco-friendly, low cost/km, less maintenance, quiet, instant torque
Challenges	High upfront cost, range anxiety, slow charging, battery life, infrastructure
